

Shoulder Arthroscopy

Rotator cuff, instability and labral injuries: how they are treated and how recovery unfolds.

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THE ROTATOR CUFF

Why the cuff matters

Four muscles and tendons (supraspinatus, infraspinatus, teres minor and subscapularis) wrap the head of the humerus and hold it centred in a shallow socket while allowing an enormous range of motion.

WHY TEARS HAPPEN

Acute tears

A fall onto an outstretched arm, a heavy lifting injury, or a dislocation, especially over the age of 40.

Degenerative tears

Gradual wear from age, repetitive overhead loading and reduced tendon blood supply.

TYPICAL SYMPTOMS

Pain

With overhead activity, and at night when lying on the affected side.

Weakness

With overhead lifting or external rotation, and difficulty reaching behind the back.

Why a torn cuff often needs repair. Unlike many soft-tissue injuries, a torn cuff tendon does not reliably heal itself: the torn end retracts and loses its blood supply. Left alone, a tear tends to enlarge, and the muscle can undergo fatty infiltration, a change that is essentially irreversible once advanced. Repairing earlier, while tendon quality and muscle bulk are still good, gives a much more reliable outcome.

How an untreated cuff injury can progress

- 1** **The shoulder stops moving normally**
Pain, guarding and stiffness
Persistent pain makes patients protect the arm. Together with inflammation, this guarding can let the capsule gradually thicken and tighten.
- 2** **Active and passive motion both restrict**
Secondary frozen shoulder
Capsular tightening produces adhesive capsulitis. In frozen shoulder both active and passive movement are restricted; in an isolated cuff tear passive range is usually preserved. Diabetes, thyroid disease and prolonged immobilisation raise the risk.
- 3** **Years of an untreated massive tear**
Cuff tear arthropathy
The humeral head loses the cuff's centring pull and migrates upward against the acromion, damaging cartilage and bone. Advanced cases develop pseudoparalysis: the deltoid fires, but without a working cuff the arm cannot be actively lifted above shoulder height.
- 4** **When the cuff cannot be restored**
Reverse shoulder replacement
A specially designed implant swaps the normal ball-and-socket anatomy (ball on the glenoid, socket on the humerus), letting the (almost always healthy) deltoid take over the cuff's job and restoring the ability to lift the arm.

JOINT REPLACEMENT

When is reverse shoulder replacement indicated?

Generally reserved for older patients, classically over roughly 65 to 70, though increasingly used in carefully selected younger patients with irreparable pathology. The priority is dependable pain relief and functional overhead motion for daily life, not high-demand athletic use.

Generally indicated for

Cuff tear arthropathy

Established cuff tear combined with significant joint arthritis.

Failed previous cuff repair

Ongoing pain and dysfunction after an earlier repair.

Massive, irreparable tears

Causing pseudoparalysis, particularly in older, lower-demand patients.

Selected fractures

Proximal humerus fractures or non-unions where cuff repair is not realistic.

Generally not recommended when a patient still has full active elevation despite a massive tear, or when the deltoid itself is not functioning. The mechanism relies entirely on a working deltoid.

Two designs, two mechanisms

ANATOMIC

Ball on the humerus

Follows natural anatomy and depends on an intact rotator cuff to move the arm.

REVERSE

Ball on the glenoid

Reverses the geometry so the deltoid does the work of the cuff. Used when the cuff cannot be restored.

INSTABILITY

After a shoulder dislocation

A traumatic dislocation, most often a fall onto an outstretched or abducted, externally rotated arm in contact or overhead sport, commonly produces two characteristic injuries as the humeral head is forced out of the socket.

PRIMARY SOURCE OF INSTABILITY

Bankart lesion

A tear of the anteroinferior labrum away from the glenoid bone, often with stretching of the capsular ligaments.

ON THE HUMERAL HEAD

Hill-Sachs lesion

A compression fracture (dent) on the back-outer part of the humeral head, created as it impacts the front edge of the glenoid.

Together, a "bipolar" lesion. Whether a Hill-Sachs lesion is likely to re-engage the glenoid rim (making it "off-track") depends on its size and position relative to glenoid bone loss. Surgeons assess this with the "glenoid track" concept on CT or MRI, and it materially changes the surgical plan.

Recurrence risk rises with younger age, ongoing collision sport and greater glenoid bone loss.

Bankart repair and posterior capsular remplissage

THE STANDARD FIX

Arthroscopic Bankart repair

For most first-time or recurrent anterior dislocations with a repairable labrum and less than roughly 15 to 20% glenoid bone loss, the torn labrum and capsule are reattached to the front of the glenoid with small suture anchors, restoring socket depth and capsular tension.

WHEN ADDED

Remplissage

When imaging shows a significant off-track Hill-Sachs lesion, the posterior capsule and infraspinatus tendon are sutured into the defect, filling it and converting an inside, joint-engaging defect into an outside one.

Under 15–20%

glenoid bone loss: Bankart repair alone

Off-track

Hill-Sachs lesion: add remplissage

Over 20–25%

glenoid bone loss: Latarjet considered

Combining Bankart repair with remplissage has been shown to achieve recurrence rates approaching those of bone-grafting procedures such as the Latarjet, while remaining a soft-tissue arthroscopic operation. It is generally preferred over Latarjet when glenoid bone loss stays below the critical threshold (roughly 20 to 25%); Latarjet is reserved for larger glenoid bone deficits or failed prior soft-tissue repairs. These thresholds guide the decision, but your surgeon weighs them alongside your age, sport, activity demands and prior surgical history.

LABRAL INJURY

SLAP tears

A SLAP tear (Superior Labrum, Anterior to Posterior) injures the top of the labrum at or near the attachment of the long head of the biceps. It is common in throwers, swimmers, tennis and volleyball players, or after a fall onto an outstretched arm.

Basic classification

I **Fraying**
Intact biceps anchor
Often simply trimmed.

II **Most common**
Labrum and biceps anchor detached
Creates true instability; the pattern where the repair-versus-tenodesis decision matters most.

III **Bucket-handle**
Intact biceps anchor
The unstable fragment is trimmed.

IV **Bucket-handle**
Extends into the biceps tendon
The tear continues into the biceps tendon itself.

Younger throwing athletes with a true Type II tear often do well with arthroscopic repair. In older, less athletically demanding patients, biceps tenodesis (releasing and re-fixing the biceps anchor lower down) frequently gives more reliable pain relief with a simpler recovery.

Physiotherapy and conservative management

Not every cuff tear needs immediate surgery. Small, partial-thickness or chronic degenerative tears in lower-demand or older patients, particularly when active motion and strength remain reasonable, are often managed first with a structured non-operative programme.

1. Settle the pain

Activity and load modification.

2. Strengthen

Scapular stabiliser and rotator cuff exercises.

3. Retrain mechanics

Posture and shoulder-blade control.

4. Graded return

Gradual return to overhead activity with regular reassessment.

Frozen shoulder needs a different pace. The initial emphasis is pain control (which may include an injection) with gentle, consistent stretching through the painful and stiffening phases, progressing to more assertive stretching in the "thawing" phase. Aggressive early stretching of a truly frozen shoulder can sometimes aggravate symptoms, so pacing is guided by which phase the shoulder is in.

RECOVERY

Post-operative rehabilitation

A repaired tendon or labrum needs time to heal biologically onto bone. Recovery is deliberately conservative in its early phases, whichever procedure was performed.

Rotator cuff repair

WEEKS 0–6	PROTECTION Sling and passive motion Support from the sling; only passive or assisted motion by the therapist, protecting the healing repair.
WEEKS 6–12	ACTIVE-ASSISTED Sling weaned, motion progresses Passive to active-assisted then active range; gentle scapular and light cuff activation begins.
3–4.5 MONTHS	STRENGTHENING Resistance work begins Resistance-band and light weighted strengthening for the cuff and scapular stabilisers.
4.5–6+ MONTHS	RETURN Sport and overhead work Overhead sport or heavy manual work is not cleared until criteria are met, often 5 to 6 months.

Bankart repair (± remplissage)

WEEKS 0–4	PROTECTION Sling worn continuously Except for hygiene and exercises. External rotation and abduction beyond safe limits are avoided; gentle pendulum motion only.
WEEKS 4–6	MOTION RESTORATION Sling gradually weaned Active-assisted motion within protected limits, guarding against excess early rotation.
WEEKS 6–12	STRENGTHENING Restoring full function Progressive cuff and scapular strengthening; full functional range of motion.
4–6+ MONTHS	RETURN TO SPORT Sport and contact progression Return to collision or overhead sport is generally not cleared before roughly 5 to 6 months.

When remplissage is added, external rotation is typically restricted a little more conservatively in the early weeks, since the procedure involves the infraspinatus tendon and posterior capsule directly.

These timelines follow the general phase-based structure of criterion-based sports medicine programmes. Always follow the specific protocol given by your own team.

QUESTIONS

What patients ask most

Short answers to the questions that come up most often before and after shoulder surgery.

Cuff tear or frozen shoulder: how do I know?

A useful clue is passive motion. In a cuff tear someone else can usually still move your arm through a nearly full range; in frozen shoulder, even passive movement is restricted.

Why operate soon rather than wait?

Tears tend to enlarge and the muscle can undergo fatty degeneration over time. Earlier repair generally gives more reliable healing.

Will every large tear need a reverse replacement?

No. Many large and even massive tears remain repairable, especially in younger patients with good tissue quality.

Does a SLAP tear always need surgery?

No. Many low-grade tears respond to physiotherapy, while higher-grade tears in younger athletes may need repair or biceps tenodesis.

Can a small cuff tear heal on its own?

Tendon tears do not reliably heal spontaneously because the torn edge retracts and loses blood supply, though small, low-grade partial tears can remain stable with physiotherapy.

Total versus reverse replacement?

An anatomic replacement relies on an intact cuff; a reverse replacement lets the deltoid move the arm without a working cuff.

Will my shoulder dislocate again after Bankart repair?

A well-selected repair (with remplissage when indicated) substantially reduces recurrence risk, though no procedure eliminates it entirely.

When can I return to overhead sport or throwing?

Generally not before about 5 to 6 months, and only once formal strength and motion criteria are met. Throwers follow a specific interval throwing programme.

Also worth asking your surgeon: why a remplissage was added to your plan, how long you will need a sling, and whether instability can ever be managed without surgery in your specific case.

Recovering well is a partnership between your surgeon, your physiotherapist and you.

Understanding why a particular technique has been recommended, and committing to the structured rehabilitation that follows, gives you the best chance of a durable, high-functioning outcome. Use this guide as a starting point for a conversation with your own treating team.